

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

900 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

900 KWP GRID-TIED SOLAR PLANT

HT CATEGORY I / II AT 11 KV

Net investment ₹3.26 Cr after input tax credit, generating 13,79,700 units in year one against an energy charge of ₹7.65 per unit.

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹6.58 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	900 kWp
Year 1 generation	13,79,700 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹98.79 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹3.42 Cr
GST at 5% — recovered as ITC	₹16.29 L
Net capitalised cost	₹3.26 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹81.98 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	3,25,71,428	1,30,28,571	1,95,42,857	32,79,031
2	1,95,42,857	78,17,143	1,17,25,714	19,67,419
3	1,17,25,715	46,90,286	70,35,429	11,80,451
4	70,35,428	28,14,171	42,21,257	7,08,271
5	42,21,257	16,88,503	25,32,754	4,24,962
6	25,32,755	10,13,102	15,19,653	2,54,978
7	15,19,653	6,07,861	9,11,792	1,52,986
8	9,11,792	3,64,717	5,47,075	91,792
9	5,47,075	2,18,830	3,28,245	55,075
10	3,28,245	1,31,298	1,96,947	33,045
11	1,96,947	78,779	1,18,168	19,827
12	1,18,168	47,267	70,901	11,896
Tax shield, years 1-12				81,79,733

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

900 kWp
PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	13,79,700	98,78,652	—	1,30,28,571	(7,92,772)	1,06,71,424	1,06,71,424
2	13,72,802	1,02,22,429	5,40,000	78,17,143	4,69,455	92,12,974	1,98,84,398
3	13,65,937	1,05,78,170	5,40,000	46,90,286	13,45,955	86,92,214	2,85,76,612
4	13,59,108	1,09,46,290	5,40,000	28,14,171	19,10,784	84,95,506	3,70,72,118
5	13,52,312	1,13,27,221	5,40,000	16,88,503	22,89,965	84,97,255	4,55,69,373
6	13,45,551	1,17,21,408	5,40,000	10,13,102	25,59,159	86,22,249	5,41,91,622
7	13,38,823	1,21,29,313	5,40,000	6,07,861	27,63,812	88,25,501	6,30,17,123
8	13,32,129	1,25,51,413	5,40,000	3,64,717	29,31,241	90,80,173	7,20,97,296
9	13,25,468	1,29,88,202	5,40,000	2,18,830	30,77,888	93,70,314	8,14,67,610
10	13,18,841	1,34,40,192	5,40,000	1,31,298	32,13,675	96,86,517	9,11,54,127
11	13,12,247	1,39,07,910	5,40,000	78,779	33,44,609	1,00,23,302	10,11,77,429
12	13,05,685	1,43,91,906	5,40,000	47,267	34,74,351	1,03,77,554	11,15,54,983
13	12,99,157	1,48,92,744	5,40,000	28,360	36,05,161	1,07,47,583	12,23,02,566
14	12,92,661	1,54,11,012	5,40,000	17,016	37,38,454	1,11,32,558	13,34,35,124
15	12,86,198	1,59,47,315	5,40,000	10,210	38,75,143	1,15,32,171	14,49,67,295
16	12,79,767	1,65,02,281	5,40,000	6,126	40,15,845	1,19,46,436	15,69,13,731
17	12,73,368	1,70,76,561	5,40,000	3,676	41,60,997	1,23,75,564	16,92,89,295
18	12,67,001	1,76,70,825	5,40,000	2,205	43,10,931	1,28,19,894	18,21,09,189
19	12,60,666	1,82,85,770	5,40,000	1,323	44,65,922	1,32,79,847	19,53,89,036
20	12,54,363	1,89,22,115	5,40,000	794	46,26,211	1,37,55,904	20,91,44,940
21	12,48,091	1,95,80,604	5,40,000	476	47,92,019	1,42,48,585	22,33,93,525
22	12,41,851	2,02,62,009	5,40,000	286	49,63,563	1,47,58,446	23,81,51,971
23	12,35,641	2,09,67,127	5,40,000	171	51,41,056	1,52,86,071	25,34,38,042
24	12,29,463	2,16,96,783	5,40,000	103	53,24,713	1,58,32,070	26,92,70,112
25	12,23,316	2,24,51,831	5,40,000	62	55,14,754	1,63,97,077	28,56,67,189
26	12,17,199	2,32,33,155	5,40,000	37	57,11,404	1,69,81,751	30,26,48,940
27	12,11,113	2,40,41,669	5,40,000	22	59,14,894	1,75,86,774	32,02,35,714
28	12,05,058	2,48,78,319	5,40,000	13	61,25,465	1,82,12,854	33,84,48,568
29	11,99,032	2,57,44,084	5,40,000	8	63,43,362	1,88,60,722	35,73,09,290
30	11,93,037	2,66,39,978	5,40,000	5	65,68,841	1,95,31,137	37,68,40,427
Total	3,85,25,585	50,82,87,288	1,56,60,000	3,25,71,421	11,57,86,857	37,68,40,427	37,68,40,427

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹6.58 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹37.68 Cr
30-year generation	3,85,25,586 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 3,85,25,586 units, avoiding 31,591.0 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹3.42 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 900 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.